KPK BOARD TEXTBOOK NOTES

# Physics - Class 9

## Unit 6: Work and Energy

Complete notes in English, based on *A Textbook of Physics, Grade IX*
Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar

> **How to use these notes.** Read one section at a time, learn the green definition boxes by heart, and copy every yellow formula box into your notebook. Work through the solved numericals before you try the multiple-choice questions (MCQs) at the end. The last page is a one-page revision sheet for the night before the exam.

### What this unit covers

1. Work - definition, formula and the unit joule
2. Energy and its forms - kinetic, potential, heat, light, sound, electrical, chemical and nuclear energy
3. Kinetic energy and potential energy - formulas and derivations
4. Interconversion of energy and the law of conservation of energy
5. Major sources of energy - renewable and non-renewable sources
6. Efficiency of a system
7. Power - definition, formula and the unit watt

### After studying this unit, you should be able to

- Define work and state its SI unit.
- Solve problems using the equation for work done.
- Define energy, kinetic energy and potential energy, and state their units.
- Derive the equations for kinetic energy and potential energy.
- State the law of conservation of energy and explain the interconversion of energy with examples.
- Describe the major sources of energy, and distinguish renewable from non-renewable sources.
- Define efficiency, explain why it is always less than 100 percent, and solve related problems.
- Define power, state its SI unit, and solve problems using power, work and time.

This checklist follows the Student Learning Outcomes for Unit 6 in the textbook.

# 6.1 Work

In everyday language, any hard activity is called work - studying, standing in a queue, or holding a heavy bag. In physics, the word has a precise meaning. Work is done only when a force moves a body.

> **Work** is done when a force acts on a body and the body moves in the direction of the force. Work is equal to the product of the force and the distance moved in the direction of the force.

$$\text{Work} = \text{Force} \times \text{Distance moved in the direction of force}$$
$$W = F \times S$$

**Conditions for work to be done:**

1. A force must act on the body.
2. The body must move (there must be displacement).
3. The movement must have a component in the direction of the force.

## The unit of work - the joule

> **One joule (J)** is the work done when a force of one newton (1 N) moves a body through a distance of one metre (1 m) in the direction of the force.

So 1 J = 1 N x 1 m = 1 Nm. Work is a **scalar** quantity - it has magnitude only, no direction. Larger units used in practice are the kilojoule (1 kJ = 1000 J) and the megajoule (1 MJ = 1,000,000 J).

## When is the work done zero?

- **No displacement:** pushing a solid wall applies a force, but the wall does not move, so no work is done on the wall.
- **No force in the direction of movement:** a porter carrying a load on his head and walking on a level road does no work against gravity, because the weight acts downward while the movement is horizontal.
- **Force at right angles to movement:** the Moon moves around the Earth under gravity, but gravity acts towards the centre and the motion is along the tangent, so gravity does no work on the Moon.

> **Exam trap:** holding a heavy bag while standing still feels tiring, but the physics work done on the bag is zero, because there is no displacement.

# Work Done by a Force at an Angle

Sometimes the force does not act along the direction of movement. If a force F acts at an angle theta to the direction of motion, only the component of the force along the motion, F cos theta, does work.

$$W = F\,S\cos(\theta)$$

- If theta = 0 degrees (force along the motion): cos 0 = 1, so W = F S - the maximum work.
- If theta = 90 degrees (force perpendicular to the motion): cos 90 = 0, so W = 0.
- If theta = 180 degrees (force opposite to the motion, like friction): cos 180 = -1, so W = -F S. The work is negative - the force takes energy away from the body.

## Work against gravity

To lift a body of mass m through a height h, we must apply an upward force equal to its weight, mg. The work done is:

$$W = F \times S = mg \times h = mgh$$

This work does not disappear - it is stored in the body as gravitational potential energy (Section 6.2).

## Solved Numerical 1

**Question:** A boy pushes a box with a force of 50 N through a distance of 8 m along the floor. How much work does he do?

**Solution:** F = 50 N, S = 8 m. W = F x S = 50 x 8 = **400 J**.

## Solved Numerical 2

**Question:** A girl lifts a bag of mass 5 kg to a shelf 1.6 m high. Find the work done against gravity. (Take g = 10 m/s$^2$.)

**Solution:** W = mgh = 5 x 10 x 1.6 = **80 J**.

## Solved Numerical 3

**Question:** A man pulls a trolley with a force of 100 N at an angle of 60 degrees to the ground, for 20 m. Find the work done. (cos 60 = 0.5.)

**Solution:** W = F S cos(theta) = 100 x 20 x 0.5 = **1000 J = 1 kJ**.

# 6.2 Energy

> **Energy** is the capacity (ability) of a body to do work.

A body that can do work possesses energy, and the amount of energy it has is measured by the amount of work it can do. Energy is a scalar quantity, and its SI unit is the same as the unit of work - the **joule (J)**.

## Forms of energy

| Form | Meaning and example |
|---|---|
| Kinetic energy | Energy possessed by a body due to its motion - a moving car, flowing water, blowing wind. |
| Potential energy | Energy stored in a body due to its position or state - water in a dam, a stretched spring. |
| Heat energy | Energy obtained from hot bodies - a stove, the Sun, burning fuel. |
| Light energy | Energy carried by light - sunlight enables plants to make food. |
| Sound energy | Energy carried by vibrating objects - loudspeakers, musical instruments. |
| Electrical energy | Energy carried by moving electric charges - used by fans, bulbs and motors. |
| Chemical energy | Energy stored in substances - food, fuels, batteries. |
| Nuclear energy | Energy stored in the nucleus of an atom, released in a nuclear power plant or the Sun. |

Kinetic energy and potential energy together are called **mechanical energy**. They are the two forms studied most carefully in this unit, because their formulas are used again and again in board numericals.

> **Mechanical energy** of a body = kinetic energy + potential energy.

# Kinetic Energy

**Kinetic energy (K.E.)** is the energy possessed by a body due to its motion.

$$\text{K.E.} = (1/2)\ m\ v^2$$

where m is the mass of the body in kilograms and v is its speed in metres per second.

## Derivation

Consider a body of mass m at rest. A constant force F accelerates it to speed v over a distance S. From kinematics, $v^2 = 2\ a\ S$, so $S = v^2 / (2a)$. The work done on the body is:

$$W = F \times S = (m\ a) \times (v^2 / 2a) = (1/2)\ m\ v^2$$

All of this work appears as the energy of motion of the body, so $\text{K.E.} = (1/2)\ m\ v^2$.

## What kinetic energy depends on

- **Mass:** double the mass (same speed) and the kinetic energy doubles.
- **Speed:** double the speed (same mass) and the kinetic energy becomes **four times**, because v is squared. This is why a fast car needs a much longer braking distance.

## Solved Numerical 4

**Question:** A car of mass 1000 kg is moving at 20 m/s. Find its kinetic energy.

**Solution:** $\text{K.E.} = (1/2)\ m\ v^2 = 0.5 \times 1000 \times (20)^2 = 0.5 \times 1000 \times 400 =$ **200,000 J = 200 kJ**.

## Solved Numerical 5

**Question:** A ball of mass 0.5 kg moves at 10 m/s. Its speed is doubled to 20 m/s. How does its kinetic energy change?

**Solution:** At 10 m/s: K.E. = 0.5 x 0.5 x 100 = 25 J. At 20 m/s: K.E. = 0.5 x 0.5 x 400 = 100 J. The kinetic energy becomes **four times** (100 J).

# Potential Energy

**Potential energy (P.E.)** is the energy possessed by a body due to its position or its state (shape).

## Gravitational potential energy

When a body of mass m is raised to a height h above the ground, work mgh is done on it against gravity. This work is stored as its gravitational potential energy.

$$\text{P.E.} = mgh$$

where g = 10 m/s$^2$ (approximately) and h is measured from a chosen reference level, usually the ground, where P.E. is taken as zero.

## Elastic potential energy

A stretched or compressed spring, a drawn bow, and a stretched rubber band store energy because work was done to change their shape. When released, this stored energy can do work - the bow shoots the arrow, and the spring of a wind-up toy turns its wheels.

| Example | Energy stored |
|---|---|
| Water stored in a high dam | Gravitational potential energy |
| A stretched spring | Elastic potential energy |
| A drawn bow | Elastic potential energy |
| A book on a high shelf | Gravitational potential energy |
| A wound-up clock spring | Elastic potential energy |

## Solved Numerical 6

**Question:** A diver of mass 60 kg stands on a board 10 m above the water. Find her potential energy. (g = 10 m/s$^2$.)

**Solution:** P.E. = mgh = 60 x 10 x 10 = **6000 J = 6 kJ**.

## Solved Numerical 7

**Question:** How high must a 2 kg stone be raised so that its potential energy is 400 J? (g = 10 m/s$^2$.)

**Solution:** h = P.E. / (mg) = 400 / (2 x 10) = **20 m**.

**Remember:** potential energy depends on the reference level you choose. Always state the height from the ground (or the level given in the question).

# 6.3 Interconversion and Conservation of Energy

Energy constantly changes from one form into another. This is called the **interconversion of energy**.

| Device or event | Energy change |
|---|---|
| Electric bulb | Electrical energy to light energy and heat energy |
| Electric motor | Electrical energy to kinetic (mechanical) energy |
| Generator (dynamo) | Kinetic energy to electrical energy |
| Burning of fuel | Chemical energy to heat energy and light energy |
| Photosynthesis in plants | Light energy to chemical energy (food) |
| Hydroelectric power station | Potential energy of water to kinetic energy, then to electrical energy |

> **Law of conservation of energy:** Energy can neither be created nor destroyed. It can only be changed from one form into another. The total energy of the universe remains constant.

## A freely falling body

A ball of mass m held at height h has P.E. = mgh and K.E. = 0, so its total energy is mgh. As it falls, its height decreases so its potential energy decreases, but its speed increases so its kinetic energy increases by exactly the same amount. Just before it touches the ground, all of its energy is kinetic. At every point of the fall:

$$\text{Total energy} = \text{P.E.} + \text{K.E.} = mgh \text{ (constant)}$$

This assumes no air resistance. With air resistance, a small part of the energy changes into heat and sound, but the total energy is still conserved - it is never lost, only spread into other forms.

# Conservation of Energy - the Pendulum

A simple pendulum is a small heavy bob hanging from a string. Its swing is the clearest classroom example of energy changing back and forth while the total stays the same.

- **At the highest point (either end of the swing):** the bob stops for an instant. Its energy is all potential energy; kinetic energy is zero.
- **At the lowest point (the mean position):** the bob moves fastest. Its energy is all kinetic energy; potential energy is minimum.
- **Between the two:** the energy is part potential and part kinetic, and the sum is constant.

In a real pendulum the swings slowly become smaller. This does not break the law of conservation of energy: the bob does work against air resistance and friction at the support, so some mechanical energy changes into heat energy. The total energy, counting the heat, is unchanged.

## Solved Numerical 8

**Question:** A stone of mass 2 kg is dropped from a height of 45 m. Find its speed just before it hits the ground, using the law of conservation of energy. (g = 10 m/s$^2$.)

**Solution:** At the top: P.E. = mgh = 2 x 10 x 45 = 900 J, K.E. = 0. Just before hitting the ground: P.E. = 0, so K.E. = 900 J. Then (1/2) m v$^2$ = 900, so v$^2$ = 900 / 1 = 900, and v = **30 m/s**.

## Solved Numerical 9

**Question:** A ball of mass 0.2 kg is thrown upward with kinetic energy 40 J. How high will it rise? (g = 10 m/s$^2$.)

**Solution:** At the highest point all the kinetic energy has become potential energy: mgh = 40, so h = 40 / (0.2 x 10) = **20 m**.

> **Key idea for numericals:** when a body rises, kinetic energy changes into potential energy; when it falls, potential energy changes into kinetic energy. Set the energy lost by one form equal to the energy gained by the other.

# 6.4 Major Sources of Energy

Almost all the energy we use comes, directly or indirectly, from the Sun. Sources of energy are divided into two groups.

> **Non-renewable sources** are used up once they are consumed and cannot be replaced within a useful time - fossil fuels (coal, oil, natural gas) and nuclear fuel.

> **Renewable sources** are replaced naturally and will not run out - sunlight, wind, flowing water, biomass, geothermal heat and tides.

## Non-renewable sources

- **Fossil fuels:** coal, oil and natural gas were formed from dead plants and animals buried millions of years ago. Burning them releases chemical energy as heat, used in vehicles, factories and thermal power stations. They produce smoke and carbon dioxide, which pollute the air and add to global warming, and the world's reserves are limited.
- **Nuclear energy:** energy released from the nucleus of atoms such as uranium in a nuclear power plant. A small amount of fuel gives a very large amount of energy, but the waste remains radioactive and dangerous for a long time.

## Why we must save energy

The demand for energy grows every year while fossil fuel reserves shrink. Using energy carefully - switching off unused lights and fans, and preferring renewable sources - protects both our pockets and the environment.

# Renewable Sources of Energy

| Source | How it is used | Point to remember |
|---|---|---|
| Solar energy | Solar cells change sunlight directly into electricity; solar cookers and water heaters use its heat. | Free and clean, but not available at night or on cloudy days. |
| Hydro (water) energy | Water stored in dams falls and turns turbines connected to generators. | Pakistan's dams, such as Tarbela and Mangla, produce electricity this way. |
| Wind energy | Moving air turns windmills and wind turbines. | Useful only where winds blow strongly and steadily. |
| Biomass energy | Burning wood, crop waste and animal dung; biogas from rotting waste is used for cooking. | Common in villages; burning produces some smoke. |
| Geothermal energy | Heat from inside the Earth turns water into steam that runs turbines. | Available only in areas with hot springs and volcanic activity. |
| Tidal energy | The rise and fall of sea tides turns turbines in coastal areas. | Needs a large difference between high and low tide. |

## Solar energy in detail

The Sun is our biggest source of energy. Plants store it as chemical energy by photosynthesis. Fossil fuels are, in effect, ancient stored sunlight. A **solar cell (photovoltaic cell)** converts light energy directly into electrical energy without any moving parts. Groups of solar cells form a solar panel, which can charge batteries for use at night - the same arrangement now common on homes and schools in Khyber Pakhtunkhwa.

## Effects on the environment

- Burning fossil fuels releases carbon dioxide and other gases, causing air pollution and global warming.
- Renewable sources produce little or no pollution while running, which is why governments encourage them.
- Every source has some cost: dams change river ecosystems, and solar panels and batteries need materials that must be disposed of safely at the end of their life.

# 6.5 Efficiency

No machine converts all the energy given to it into useful work. Some energy is always wasted, mostly as heat due to friction, and as sound.

> **Efficiency** of a system is the ratio of the useful energy (or work) output to the total energy input, expressed as a percentage.

Efficiency = (Useful output energy / Total input energy) x 100%

- Efficiency has **no unit** - it is a ratio, written as a percentage.
- Efficiency is **always less than 100 percent**, because some input energy is always wasted as heat and sound.
- A system with efficiency 100 percent, which wastes nothing, is called an **ideal system**. No real system is ideal.

| System | Typical efficiency |
|---|---|
| Ordinary electric bulb (filament) | About 10% - most energy becomes heat |
| LED bulb | Much higher than a filament bulb - most energy becomes light |
| Petrol car engine | About 25-30% |
| Electric motor | About 70-90% |

The values in this table are typical approximate values for comparison, of the kind used in textbook examples.

## Solved Numerical 10

**Question:** A motor takes 2000 J of electrical energy and does 1400 J of useful work. Find its efficiency.

**Solution:** Efficiency = (1400 / 2000) x 100% = **70%**. The remaining 600 J is wasted as heat and sound.

## Improving efficiency

Lubricating moving parts reduces friction, and good maintenance keeps machines near their best efficiency. Choosing LED bulbs instead of filament bulbs is a household example of the same idea.

# 6.6 Power

Two workers may do exactly the same work, but one finishes in half the time. The faster worker is more powerful. Power measures the **rate** of doing work.

> **Power** is the rate at which work is done, or the rate at which energy is converted from one form into another.

$$\text{Power} = \text{Work done} / \text{Time taken}$$
$$P = W / t$$

## The unit of power - the watt

> **One watt (W)** is the power of a system that does one joule of work in one second: 1 W = 1 J/s.

- 1 kilowatt (kW) = 1000 W
- 1 megawatt (MW) = 1,000,000 W
- 1 horsepower (hp) = 746 W - still used for engines and motors

## The kilowatt hour - a unit of energy, not power

Electricity bills are charged in **units**, where 1 unit = 1 kilowatt hour (kWh). It is the energy used by a 1000 W appliance running for 1 hour:

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3{,}600{,}000 \text{ J} = 3.6 \text{ MJ}$$

## Solved Numerical 11

**Question:** A crane does 30,000 J of work in 10 s. Find its power.

**Solution:** P = W / t = 30,000 / 10 = **3000 W = 3 kW**.

## Solved Numerical 12

**Question:** How much energy in kWh does a 100 W bulb use in 10 hours?

**Solution:** Energy = Power x Time = 0.1 kW x 10 h = **1 kWh = 1 unit**.

> **Do not mix them up:** the watt measures power (how fast energy is used); the joule and the kilowatt hour measure energy (how much is used in total).

# MCQs - Choose the Correct Answer

1. The SI unit of work is: (a) newton (b) joule (c) watt (d) pascal
2. Work is done when: (a) a force acts but the body does not move (b) a body moves with no force on it (c) a force moves a body in its own direction (d) a body is held still
3. A force of 10 N moves a body 5 m. The work done is: (a) 2 J (b) 15 J (c) 50 J (d) 0.5 J
4. The work done by a force acting at right angles to the motion is: (a) maximum (b) zero (c) negative (d) equal to F S
5. The energy of a body due to its motion is: (a) potential energy (b) heat energy (c) kinetic energy (d) chemical energy
6. Kinetic energy is calculated by: (a) mgh (b) (1/2) m $v^2$ (c) F x S (d) W / t
7. If the speed of a body is doubled, its kinetic energy becomes: (a) double (b) half (c) four times (d) the same
8. Water stored in a dam has: (a) kinetic energy (b) potential energy (c) sound energy (d) nuclear energy
9. The law of conservation of energy says energy: (a) can be created (b) can be destroyed (c) changes form but total stays constant (d) always decreases
10. Which of these is a renewable source? (a) coal (b) natural gas (c) sunlight (d) diesel
11. Efficiency is always: (a) more than 100% (b) equal to 100% (c) less than 100% (d) zero
12. The SI unit of power is: (a) joule (b) watt (c) newton (d) volt
13. 1 horsepower is equal to: (a) 100 W (b) 746 W (c) 1000 W (d) 3600 W
14. 1 kWh is equal to: (a) 3.6 kJ (b) 36 kJ (c) 3.6 MJ (d) 36 MJ

Answer key is on the last page, with the revision sheet. Try every question yourself first.

# One-Page Revision Sheet

## Formulas to learn by heart

$$W = F \times S \quad | \quad W = F\,S\cos(\theta) \quad | \quad K.E. = \tfrac{1}{2} m v^2$$
$$P.E. = mgh \quad | \quad P = W / t \quad | \quad \text{Efficiency} = (\text{Useful output} / \text{Total input}) \times 100\%$$

## Key points

- Work needs a force AND displacement in the direction of the force. No displacement, no work.
- 1 joule = work of 1 newton acting through 1 metre. Work and energy are both measured in joules.
- Energy is the capacity to do work. Kinetic energy depends on mass and on the square of speed.
- Potential energy mgh is measured from a reference level, usually the ground.
- Energy is never created or destroyed; it only changes form. P.E. + K.E. stays constant for a falling body (without air resistance).
- Non-renewable sources (coal, oil, gas, nuclear fuel) run out; renewable sources (solar, wind, water, biomass, geothermal, tidal) do not.
- Efficiency is always below 100% because of friction, heat and sound. It has no unit.
- Power is the rate of doing work. 1 W = 1 J/s; 1 hp = 746 W; 1 kWh = 3.6 MJ is a unit of energy.

## MCQ answer key

1 (b) 2 (c) 3 (c) 4 (b) 5 (c) 6 (b) 7 (c) 8 (b) 9 (c) 10 (c) 11 (c) 12 (b) 13 (b) 14 (c)

## Self-test before the exam

1. Define work and state the condition under which a force does no work.
2. Derive K.E. = (1/2) m $v^2$.
3. Prove that the total energy of a freely falling body remains constant.
4. Write three differences between renewable and non-renewable sources of energy.
5. A 1500 W heater runs for 2 hours. How many units (kWh) of energy does it use? *(Answer: 3 kWh.)*